

# MALIK JIRASEK

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Institute for Theoretical Physics  
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## EDUCATION

Oct 2025 – Present

Master of Science in Physics, University of Tübingen, Germany

Thesis: "Keldysh-field-theory approach to the quantum Fisher information"

Supervisor: Igor Lesanovsky

Aug 2020 – Jul 2025

Bachelor of Science in Physics, University of Tübingen, Germany

Final grade: 1.04 / 5.0 (German system; 1.0 = excellent)

Thesis: "Quantum Sensing beyond the Heisenberg Limit with the Boundary Time Crystal"

Supervisor: Igor Lesanovsky

## RESEARCH EXPERIENCE

Nov 2025 – Present

Master of Science Thesis Research, University of Tübingen, Germany

- Developing a Keldysh-field-theory description of the quantum Fisher information in open quantum systems, with a focus on scalable sensor architectures.
- Demonstrating genuine quantum advantage in driven-dissipative many-body sensor architectures.

Aug 2025 – Present

Working Student, Bosch eBike Systems, Germany

- Implementing a Machine Learning pipeline for battery cell aging predictions, using BiLSTM networks and TensorFlow.
- Collaborating with the engineering team to integrate the model into the production workflow.

Oct 2024 – Jul 2025

Bachelor of Science Thesis Research, University of Tübingen, Germany

- Demonstrated sub-Heisenberg scaling of precision in optical-phase estimation using a boundary time crystal as a light source.
- Designed and numerically benchmarked realistic measurement protocols with QuTiP simulations.
- Results incorporated into a manuscript submitted to *Physical Review Letters* (under review).

Aug 2024 – Oct 2024

Research Intern, Humboldt University of Berlin, Germany

- Extended a Fortran-based many-body solver to include high-precision interaction potentials for ultracold-atom tweezer arrays, achieving  $10^{-8}$  relative precision in ground state energies.

## PUBLICATIONS

In preparation

Malik Jirasek, Igor Lesanovsky, and Viktoria Noel, "Keldysh-field-theory approach to the quantum Fisher information". *In preparation*.

Preprint / under review

Malik Jirasek, Igor Lesanovsky, and Albert Cabot, "The Boundary Time Crystal as a light source for collectively enhanced sensing", arXiv:2511.23416 [quant-ph], 2025. Accepted for publication in *Physical Review A*.

## PRESENTATIONS & CONFERENCES

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01 Oct 2026 Accepted for poster at DQUT Conference, Palma de Mallorca  
07 Sep 2026 Presented poster at QUANT26 Master School, Dresden  
06 Mar 2026 Contributed Talk at DPG Spring Meeting SAMOP, Mainz

## TEACHING EXPERIENCE

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WS 2024/25 Advanced Quantum Mechanics tutor (University of Tübingen)  
SS 2024 Quantum Mechanics tutor (University of Tübingen)  
WS 2023/24 Analytical Mechanics tutor (University of Tübingen)  
SS 2023 Physics 2 tutor (University of Tübingen)  
WS 2022/23 Physics 1 tutor (University of Tübingen)  
WS 2021/22 - WS 2024/25 Physics Lab tutor (University of Tübingen)

## AWARDS & SCHOLARSHIPS

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2021 & 2023 Deutschlandstipendium (merit-based; awarded to ~0.5% of enrolled students)

## SKILLS & METHODS

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Programming Python (TensorFlow, QuTiP), Fortran, MATLAB (Simulink)  
Simulation & Modeling Lindblad formalism, Keldysh formalism, stochastic calculus  
Tools Linux, Git, LaTeX  
Languages German (native), English (full professional proficiency), French (beginner)

## SERVICE & LEADERSHIP

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Oct 2026 Co-organizer, Workshop "Nonequilibrium methods in driven-dissipative many-body systems"  
2026 Student representative, faculty appointment committee (ML in physics)  
2022 - 2025 Treasurer & member, physics student council

## REFERENCES

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Available on request.